Cardiac

1. Adrenaline
2. Adenosine
3. Amiodarone
4. Procainamide (not available in SA)
5. Atropine Sulphate
6. Dopamine
7. Dobutamine
8. Lignocaine
9. Magnesium Sulphate

ACS

1. Aspirin
2. Clopidogrel
3. Glyceryl Trinitrate
4. Streptokinase
5. Tenecteplase
6. Enoxaparin
7. Heparin Sodium

Anti-Hypertensives

1. Hydralazine
2. Labetalol
3. Nifedipine
4. Sotalol (not in ECP scope)

Diuretic

1. Furosemide

OBS

1. Betamethasone
2. Dexamethasone
3. Oxytocin
4. Magnesium Sulphate

Respiratory

1. Hydrocortisone
2. Methylprednisole
3. Prednisolone
4. Ipratropium Bromide
5. β2 Stimulants
6. Magnesium Sulphate

Miscellaneous

1. Tranexamic Acid
2. Medical oxygen

Antidote/ Overdose

1. Activated Charcoal
2. Flumazenil
3. Naloxone
4. Promethazine

Electrolyte/ Metabolic

1. Calcium chloride/ gluconate
2. Sodium Bicarbonate 8.5%
3. Dextrose
4. Oral Glucose
5. Glucagon
6. Thiamine

Analgesics

1. Fentanyl
2. Morphine
3. Nitrous oxide
4. Paracetamol
5. Penthroxyflurane

Sedative/ Induction

1. Diazepam
2. Lorazepam
3. Midazolam
4. Etomidate
5. Ketamine

Paralytics

1. Rocuronium
2. Suxamethonium
3. Vecuronium (Not in ECP scope)
4. Sugammadex
5. Neostigmine

Anti-emetics

1. Ondansetron
2. Metoclopramide

Adrenaline

Adrenoceptor agonist/ Sympathomimetic



α-effects

- Vasoconstriction (arterial)
- Increased breakdown of glycogen
- Mydriasis
- Diaphoresis/ Piloerection
- Sphincter constriction

β1-effects

- Positive inotrope
- Positive chronotrope
- Positive dromotrope
- Increased myocardial O2 consumption

β2-effects

- Smooth muscle relaxation (bronchial, vascular, bladder, intestinal, uterine) → Venodilation
- Breakdown of glycogen to glucose

Onset of action: Immediate
Duration of action: 10-15 minutes

Contra-indications

- Tachyarrhythmias
- Pheochromocytoma
- Hypersensitivity to sulphites

Precautions

- Do not mix with alkaline solutions
- Hypertension
- Diabetes mellitus
- Hyperthyroidism
- Pregnancy

Packaging

1mg/1ml (1:1000)/ 0.1%
Diluted in 9ml saline = 0.1mg/ml

Dosage and administration

CARDIAC ARREST

Adult	Child
1mg IVI push Repeat every 3-5 min Infusion post ROSC (SBP<70mmHg): 2-10ug/min → add 1mg to 200ml NS = 5ug/ml	0,01mg/kg IVI push Repeat every 3-5 min Infusion post ROSC: 0,1ug/kg/min → add 1mg to 200ml NS = 5ug/ml

RESISTANT SYMPTOMATIC BRADYCARDIA

Adult	Child
SBP< 85mmHg 2-10ug/min - Titrate	0.01mg/kg IVP every 3 min Infusion: 0,1-1ug/kg/min

ANAPHYLAXIS

Adult	Child
IM: 0.3mg undiluted Repeat every 15-20 min IV: 0.1mg IVI diluted slowly over 5 min	IM: 0.01mg/kg every 5-15 min undiluted IV: 0.01mg/kg diluted slowly

NEAR FATAL ASTHMA

Adult	Child
SC/IM: 0.3mg every 20 min (3 doses)	0.01mg/kg every 20 min (max 3 doses)

CROUP/ LARYNGEAL EDEMA

Adult & Child
Nebulized with 1ml of 1:1000 + 4ml NS
If necessary, increase to 2-4mg in 5mlNS

Adenosine

Supraventricular anti-arrhythmic; Endogenous purine nucleoside



Pharmacological Action

- Conduction in SA and AV node → Slows inward Ca2+ flow and therefore decreases automaticity and rate of discharge. Adenosine is produced from the de-phosphorylation of ATP. It blocks the influx of Ca2+ by inhibiting cAMP formation. This therefore terminates supraventricular arrhythmias due to re-entry pathways involving the AV node.
- SVTs that do not involve the AV node (eg WPW syndrome) are usually not terminated by adenosine; especially true for atrial flutter/ fibrillation.
- Acting on A2B2 receptors in the lungs results in mast cell degranulation leading to bronchoconstriction.

Adverse Effects

- CVS → Vasodilation; flushing; light-headedness; headache; hypotension
- Arrhythmias → Chest pain; palpitations
- Other → Paresthesia; diaphoresis; dyspnoea; bronchospasm; nausea; metallic taste in mouth

Dosage and Administration

6mg rapid IV push – followed immediately by 20ml NS IVP flush (Draw up drug and flush in 2 syringes)
If no response after 2 min → 12 mg rapid IVP
A second 12mg dose can be considered if no response

Packaging

6mg/2ml glass vial

Note

Adenosine is administered extremely rapidly using the 2 syringe technique

Onset of action: Immediate
Duration of action: 1-2 min

Indications

Stable patients with narrow-complex paroxysmal SVT, to terminate re-entry SVT.

Contra-indications

- Allergy to adenosine
- 2nd/ 3rd degree heart block
- Sick sinus syndrome
- Drug-induced tachycardia
- Atrial flutter/ fibrillation
- Transplanted hearts
- WPW syndrome
- Unstable asthmatics

Precautions

- Can induce arrhythmias (tachy/brady)
- Avoid in patients taking dipyramole, carbamazepine (need to halve dose if required)
- Less effective in patients taking theophylline (may need to double usual dose)
- May cause bronchoconstriction in asthmatics
- Patient must be supine during administration

Amiodarone

Antiarrhythmic



Packaging
150mg/3ml ampoule

Pharmacological Action

Amiodarone (an iodine-containing agent) prolongs the action potential of the cardiac cell, whilst simultaneously blocking sodium, potassium and calcium channels, therefore delaying repolarization, and blocking further excitation. It also blocks alpha and beta adrenergic receptors thus rendering free floating catecholamines ineffective at causing excitation.

Indications

- Defibrillation refractory – VF and refractory pulseless VT (ie, unresponsive to CPR, shock and vasopressor)
- Control of hemodynamically stable VT
- Polymorphic VT with normal QT interval (ie, excluding Torsades de pointes)
- Narrow-complex re entry SVT if uncontrolled by adenosine and vagal manoeuvres, or when these are contra-indicated

Dosage and Administration

CARDIAC ARREST

- Administer 300mg IVI rapid bolus
- If VF or pVT does not respond to defibrillation, initial amiodarone dose, and further defibrillation, administer an additional dose of 150mg in 3-5 mins
- Infusion: Following successful defibrillation with Amiodarone, a slow IV infusion of 1mg/min may be administered
- Maximum cumulative dose: 2.2g IV/ 24 hours

STABLE VT/SVT

- 150mg IVI over 10 min (15mg/min)
- Infusion: slow IVI of 1mg/min

Contra-indications

- SA or AV block
- Sinus bradycardia
- Allergy to iodine
- Prior use of Lignocaine
- Torsades de pointes

Precautions

- Never to be used with or following Lignocaine
- **Should be diluted with 5% Dextrose (D5W)**

Adverse Effects

- Vasodilation/ hypotension
- Negative inotrope
- Negative chronotrope
- Prolongation of QT interval

Onset of action: 10-15 min
Half-life: 40 days

Atropine

Anti-cholinergic/ Anti-muscarinic

Onset of action: Immediate
Half life: 2-4 hours



Pharmacological Action

- Atropine acts as a competitive antagonist at muscarinic (cholinergic) receptor sites, blocking the stimulation of parasympathetic nerve fibers
- Heart: Supraventricular conductive tissue → positive chronotrope; positive dromotrope
- Eyes → mydriasis
- Exocrine glands → decreased sweat, tears, salivary and pancreatic excretions
- Lungs → reduction in bronchial secretions; some bronchodilation
- Digestive system → decreased peristalsis; sphincter constriction
- Urinary → bladder relaxation; sphincter constriction

Precautions

- First rule out other causes of bradycardia (hypoxia, hypothermia, raised ICP, hyperkalemia)
- 2nd degree type II and 3rd degree AV blocks with wide QRS complexes → Atropine may induce paradoxical bradycardia
- Ischemic heart disease, hypertension, heart transplant patients
- Toxins (eg, Beta blocker OD, calcium channel blocker OD)

Indications

- Symptomatic bradycardia, associated with unstable signs or symptoms (acute decreased LOC, ischemic chest pain, hypotension, pulmonary edema, CCF)
- Bradycardia associated with ventricular extra systoles
- Asystole/ PEA with bradycardia
- Organophosphate poisoning

Packaging

Glass/ plastic ampoules:
0.5mg/1ml
0.6mg/1ml
1.0mg/1ml
1.2mg/1ml

Adverse Effects

- Confusion/ restlessness
- Diplopia, mydriasis, photophobia, blurred vision
- Dry mucous membranes
- Tachycardia, palpitations, arrhythmias, paradoxical bradycardia
- Dry, hot flushed skin
- Acute urinary retention
- Constipation, nausea, vomiting

DOSAGE AND ADMINISTRATION

ASYSTOLE/ PEA BRADYCARDIA		REFRACTORY SYMPTOMATIC BRADYCARDIA		ORGANOPHOSPHATE POISONING	
Adult	Child	Adult	Child	Adult	Child
1mg IVI push every 3-5 min; Maximum – 3mg	Not indicated	0.5mg IVI push every 3-5 min; Maximum – 3mg	0.02mg/kg IVI push; repeat once after 3-5 min; Maximum – 0.04mg/kg	0.5-2.5mg IVI push every 4 min until atropinisation (decreased secretions)	0.02mg/kg IVI push every 4 min until atropinisation (decreased secretions)

Dopamine

Sympathomimetic; Inotrope; Vasopressor



Pharmacological Action

Chemical precursor of norepinephrine that stimulates dopaminergic, beta-2-adrenergic, and alpha-adrenergic receptors. Dosage-related effects:

- 1-2ug/kg/min produces vasodilation of renal, mesenteric, and cerebral arteries.
- 2-10ug/kg/min stimulates both beta-1- and alpha-adrenergic receptors, resulting in increased cardiac output.
- > 10ug/kg/min stimulates alpha-adrenergic receptors, resulting in renal, mesenteric, and peripheral arterial and venous vasoconstriction.

Indications

Cardiogenic shock and hemodynamically significant hypotension

Contra-indications

- Pheochromocytoma (causes serious acute hypertension)
- Hypovolemia related to severe trauma

Onset of action: 5 minutes
Half-life: <10 minutes

Dosage and Administration

5ug/kg/min titrated to SBP>90mmHg

Packaging

200mg/5ml vial

Adverse Effects

- Tachydysrhythmias
- Ectopic beats
- Nausea and vomiting

Precautions

- Patients receiving monoamine oxidase inhibitors should receive no more than 1/10 normal dose
- Dopamine should not be discontinued abruptly but should be tapered gradually

Dobutamine

Synthetic catecholamine

Onset of action: 1-2 min
Infusion takes 10 min to reach peak effects

Contra-indications

- Hypovolaemia
- Idiopathic hypertrophic subaortic stenosis
- Allergy to sulphites



Packaging

250mg/20ml vial

Pharmacological Action

- Dobutamine is a synthetic catecholamine and a direct acting inotrope.
- It stimulates the beta receptors in the heart and produces mild chronotropic, hypertensive, arrhythmogenic and vasodilative effects.
- β-1 stimulation results in an increase in heart rate (mild), myocardial contractility and excitability.
- β-2 stimulation is minimal – may have some peripheral vasodilatation and brochodilation.

Precautions

- In patients who are given beta-adrenergic receptor agonists the potency of dobutamine may be decreased. This could potentiate the alpha agonist effects such as hypertension and vasoconstriction. On the contrary, alphaadrenergic blockade may make the beta1 and beta 2 effects apparent, resulting in tachycardia & vasodilatation.
- Serum potassium levels should be monitored as dobutamine like other beta-agonists can produce mild reduction in potassium concentration
- Hypovolemia should be corrected before treatment with dobutamine is commenced.

Indications

- Acute heart failure
- Cardiogenic shock
- To reduce preload and afterload in cardiogenic pulmonary oedema.
- To increase cardiac output, improve contractility and oxygen delivery.

Adverse Effects

- Tachydysrhythmias and hypertension
- Hypotension
- Mild reduction in serum potassium concentrations

Dosage and Administration

- Reconstitute vial with 10 mLs of water for injection, 0.9% sodium chloride should not be used to reconstitute vial but may be used for further dilution
- Add 250mg dobutamine to 40mLs sterile 0.9% sodium chloride, to give a final volume of 50ml and final concentration of 5mg/mL, or 5000ug/mL
- Desired dose range is 2.5 - 15ug/kg/minute

Lignocaine

Ventricular anti-arrhythmic



Pharmacological Action

- Conduction in SA and AV node has slow Ca2+ flow. All other myocardial tissues have fast inward Na+ flow. Lignocaine blocks Na+ and K+ ions in ischemic cells and thus mainly affect ventricular conduction.
- Acts as a membrane stabilizer → inhibition of fast Na+ channels; termination of ectopic beats; shortened action potential duration; decreased myocardial excitability.
- In emergency intubation, Lignocaine mitigates catecholamine effects of pharyngeal/ laryngeal manipulation in patients with head injury, stroke, intracerebral bleeds or any situation where raised ICP, intra-ocular pressure of BP will be detrimental.

Packaging

- 50mg/5ml (1%) ampoule
- 100mg/5ml (2%) ampoule
- 500mg/5ml (10%) ampoule
- 1000mg/5ml (20%) ampoule

Indications

- Shock resistant VF and VT (only if amiodarone unavailable)
- Control of hemodynamically compromising PVCs – complex ventricular ectopy (symptomatic/ unstable patient) associated with ACS/ AMI
- Torsades the pointes (only if MgSO4 ineffective/ unavailable)
- Intubation of patients where increased ICP is detrimental

Adverse Effects

Early signs of toxicity include numbness of tongue and peri-oral region

Toxic effects: (1) CNS excitation (restlessness, muscle twitching, seizures); (2) CNS depression (drowsiness; respiratory failure; coma); (3) CVS depression (hypotension; bradycardia; cardiac arrest).

Contra-indications

- Prior use of Amiodarone
- 2nd/ 3rd degree heart blocks
- Bradycardia
- Hypotension not due to ventricular arrhythmia
- Severe sinus node dysfunction
- Accelerated idioventricular rhythm

Dosage and administration

STABLE VT/ COMPLEX VENTRICULAR ECTOPY

Adult

Loading dose 1mg/kg slowly IVI → Repeat loading dose 0.5mg/kg every 5 min; Max dose = 3mg/kg. Maintenance infusion: 1-4mg/min (20-50ug/kg/min) upon restoration of stable a rhythm.

UNSTABLE VT

Adult

- Synchronized cardioversion → loading dose and maintenance infusion (as above) once cardioversion is successful
- If VT recurs after successful cardioversion, cardiovert again using last successful joules
- Once successful, administer further bolus of 0.5mg/kg lignocaine and continue infusion.

VF (IF NO AMIODARONE)

Adult

1mg/kg IV/IO followed by 0.5mg/kg IV/IO every 5 min → successful defibrillation → maintenance infusion.

INTUBATION (RAISED ICP)

Adult

1.5mg/kg 3 min before intubation

Children doses same as adult except infusion is 20-50ug/kg/min.

Onset of action: Immediate

Duration of action: 10-20 minutes

Magnesium Sulphate

Mineral



Packaging
1g/2ml ampoule

Pharmacological Action

Mg is essential for movement of Na+, K+ and Ca2+ in and out of the cells in the body → plays important role in stabilizing excitable membranes. It prevents seizures by blocking neuromuscular transmission. Acts peripherally, causing vasodilation leading to hypotension with higher doses. Hypokalemia combined with hypomagnesia may lead to dysrhythmias, including VF.

Dosage and Administration

REFRACTORYCARDIAC ARREST/ ACUTE SEVERE ASTHMA

Adult	Child
1-2g (2-4ml of 50% solution) → Dilute 1 vial to 10ml with sterile water for 10% solution → administer slowly, not exceeding 1.5ml/min	25-50 mg/kg given over 10-20 min (Max dose 2g)

SEIZURES IN TOXEMIA OF PREGNANCY

Bolus	Infusion
2-4g of 10% solution administered slowly, not exceeding 1.5ml/min (over 10-20 min)	3-4g in 200ml NS infused @ 50ml/hour; Alternative – 5g MgSO4 deep IM in each buttock

Indications

- Acute severe asthma unresponsive to conventional therapy
- VF or pVT associated with torsades de pointes
- Prevention/ control of seizures in pre-eclampsia/ eclampsia

Contra-indications

- Heart block or myocardial damage
- Nebulized MgSO4 not recommended for adults with acute asthma exacerbation

Precautions

- Rapid administration can lead to severe hypotension
- Use with caution in renal failure
- MgSO4 crosses the placenta – may produce respiratory depression in the neonate when used within 2 hours prior to delivery

Adverse Effects

- CNS → Drowsiness; depressed reflexes; flaccid paralysis
- CVS → Hypotension; flushing; circulatory collapse
- Metabolic → Hypocalcemia
- Respiratory → Respiratory paralysis
- Skin → Diaphoresis
- Hypothermia

Onset of action: IV = 1-2 min; IM = 1 hour
Duration of action: IV = 1-2 hours; IM = 3-4 hours

Aspirin

Non-steroidal anti-inflammatory/ platelet aggregation inhibitor

Pharmacological Action

Aspirin inhibits the enzyme cyclo-oxygenase thus inhibiting the production of prostaglandins including thromboxane; it has no effect on leukotriene production.

Indications

Suspected myocardial infarction

Contra-indications

- Aspirin allergy
- Peptic ulceration with active bleeding
- Bleeding tendency
- Patients already receiving platelet aggregation inhibitors or anticoagulants
- Pregnancy
- Children < 12 years old
- Severe renal impairment/ kidney transplant

Adverse Effects

- Anaphylaxis – some patients, especially asthmatics exhibit notable sensitivity to aspirin, which my provoke hypersensitivity reactions
- Potential bronchospasm in asthmatics
- Gastric mucosal irritation (dyspepsia; peptic ulceration/ bleeding)
- Bleeding tendency
- Suppression of uterine contractions

Precautions

- Bronchial asthma
- Patient must be conscious

Dosage and Administration

Administer 150-300mg orally, chewed, crushed or dissolved → Do not use high 500mg dose; do not use enteric coated aspirin

Packaging

Regular aspirin: 300mg tablets
Extra strength: 500mg tablets
Dispersible aspirin: 100mg & 300mg tablets



Clopidogrel

Platelet aggregation inhibitor

Pharmacological Action

- Intererence with ADP binding to the platelet receptor (P2Y12), and ultimate ADP mediated activation of the glycoprotein complex (GPIIb/IIIa) leads to irreversible, non-competitive inhibition of platelet aggregation, without influencing cyclo-oxygenase.
- By inhibition of platelet aggregation induced by other agonists by blocking amplification of platelet activation by released ADP.

Precautions

- Increased risk of bleeding in trauma/ surgery/ pathological conditions
- Liver/ renal impairment
- NSAIDs – Increased risk of GIT bleeding

Dosage and Administration

300mg loading dose in ACS < 75 years of age and 75mg if > 75 years old – Mandatory in combination with fibrinolysis

Indications

- Anti-platelet therapy in true aspirin allergy in patients with suspected AMI
- Prior to primary PCI and suggested for pre-hospital fibrinolysis

Adverse Effects

- Bleeding (common)
- GIT → abdominal pain, dyspepsia, diarrhoea, nausea
- CNS → Headache, dizziness, vertigo, paraesthesia
- Skin rash and pruritis
- Rare → neutropenia, thrombotic thrombocytopenic purpura, anaphylaxis

Contra-indications

- Known allergy
- Active bleeding eg,peptic ulcer, intracranial hemorrhage

Onset of action: 2 hours
Duration of action: 5 days

Packaging

75mg tablets



Glyceryl Trinitrate

Vasodilator



Pharmacological Action

Nitrates cause dilation of the venous system, which decreases venous return (preload) and decreases myocardial wall tension. This improves sub-endocardial perfusion. Nitroglycerine dilates the coronary arteries, antagonizes coronary vasospasm and increases coronary collateral blood flow to the ischemic myocardium

Precautions

- Patient must be positioned sitting/ semi-fowlers prior to administration
- Do not administer simultaneously to other vasodilators
- Do not shake the aerosol prior to administration

Onset of action: 1-3 minutes
Duration of action: 30-60 minutes

Contra-indications

- Hypotension (SBP < 90mmHg)
- Decrease in normal BP > 10% or 30% if hypertensive
- Sildenafil (Viagra) or Vardenafil (Levitra) taken during preceding 24 hours
- Tadalafil taken in last 48 hours
- Bradycardia/ severe tachycardia
- Right ventricular infarction

Adverse Effects

- Headache
- Hypotension
- Tachycardia

Dosage and Administration

One sublingual tablet or one sublingual spray (without inhaling) → Repeat every 5 min until pain relieves (Max dose = 3 sprays/ 3 tablets)

Indications

- Angina pectoris
- AMI
- Acute pulmonary edema

Packaging

- Nitrolingual spray containing 200 x 0.4mg atomized sprays
- 0.5mg tablets, in amber coloured container

Streptokinase

Thrombolytic

Pharmacological Action

Has no enzymatic activity of its own. Streptokinase indirectly activates plasminogen by forming a high affinity complex with plasminogen or plasmin. The formed streptokinase-plasminogen complex converts other plasminogen molecules to plasmin which in turn degrades fibrin thrombi and fibrinogen.

Precautions

- Increased risk of bleeding
- CVS Effects → Tachycardia/ bradycardia; dysrhythmias; severe symptomatic hypotension.

Adverse Effects

- Anaphylaxis
- Back pain
- Chills
- Haemorrhage
- Headache
- Hypotension
- Pyrexia
- Rashes

Contra-indications

Absolute

- Streptokinase use < 1 year
- CVA < 1 year
- Severe haemorrhage < 1 month
- Recent major trauma
- Bleeding disorders
- Brain/ spinal surgery < 1 month
- Endocarditis

Relative

- Refractory hypertension (>180/110)
- Warfarin therapy
- Laser treatment of retina
- Pregnancy/ recent delivery/ abortion
- TIA < 6 months
- Impaired liver/ kidney function

Dosage and Administration

- Reconstitute → Inject 5ml NS into vial; Gently invert vial to reconstitute; Withdraw entire volume from vial and dilute further into 200ml NS.
- Adult infusion: 1 500 000 IU in 200ml NS over 60min.

Packaging

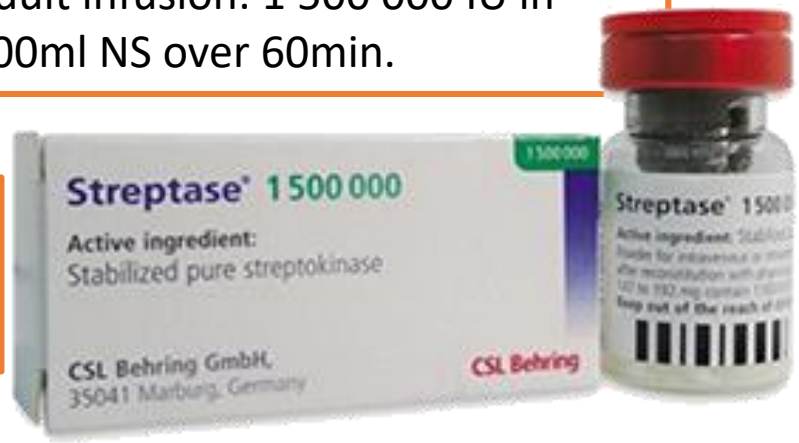
1 500 000 IU/ 750 000 IU lyophilized white powder in clear glass vial

Indications

AMI with ST-elevation in adults if onset < 6 hours

Onset of action: Immediate

Duration of action: 4-12 hours



Tenecteplase

Thrombolytic

Pharmacological Action

Tenecteplase degrades the fibrinous matrix of thrombi associated with vascular occlusion and ischemia. It is a tissue Plasminogen activator (tPa); also naturally occurs in the body. Activation of Plasminogen causes its conversion to plasmin; an enzyme which binds to and breaks down fibrin via fibrinolysis.

Indications

AMI

Onset of action: 15 min
Duration of action: Several hours



Contra-indications

- High risk of bleeding
- Currently on anticoagulants
- Previous injury to CNS (tumor, surgery, aneurysm, etc.)
- Bleeding tendency
- Severe hypertension
- Major surgery in past 2 months
- Pericarditis/ Endocarditis
- Acute pancreatitis
- Hepatic dysfunction
- Peptic ulcers
- Known history of stroke
- Dementia
- Recent head trauma/ CPR

Adverse Effects

- CVS → Haemorhage; Arrhythmias; embolism
- Immune → Anaphylaxis
- CNS → Intracranial haemorrhage
- Respiratory and GIT haemorrhage; nausea and vomiting
- Ecchymosis

Precautions

- Carefully consider risk vs benefit in terms of transport for PCI
- Carefully assess bleeding risk
- Avoid if STEMI diagnosis uncertain
- Concominant anticoagulant therapy is mandatory (Enoxaparin/ aspirin/ clopidogrel)

Dosage and Administration

- Dose based on patient weight (see table)
- Patients > 75 years should receive half normal dose
- Reconstituted with solvent; IV administration of single bolus given over 10 sec

Packaging

- Dry powder: 40mg vial (8000 units) + 8ml solvent
- 50mg (10 000 units) vial + 10ml solvent

Enoxaparin

Anticoagulant (Clexane)

Pharmacological Action

Enoxaparin is a low molecular weight heparin which has a greater antithrombotic activity (anti-factor Xa) than a thrombolytic effect (anti-factor IIa activity) *in vivo.* It is well absorbed after subcutaneous injection.

Indications

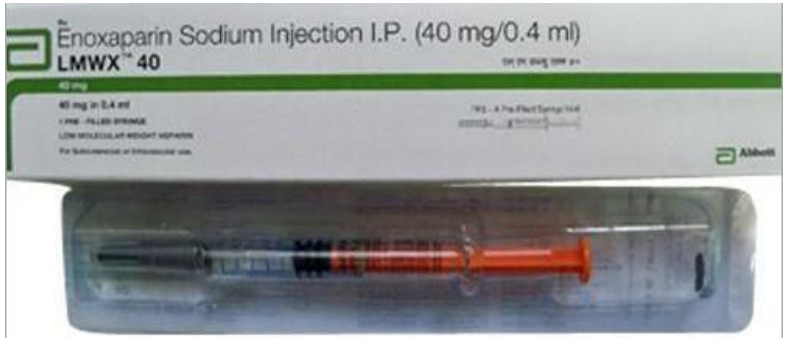
Enoxaparin is indicated as co-therapy to tenecteplase for thrombolytic therapy for acute phase of myocardial infarction by Emergency Care Practitioners.

Adverse Effects

- Haemorrhage
- Thrombocytopenia
- Local reactions: Pain, hematoma

Contra-indications

- Allergy
- Patients who are haemorrhaging (normal menstruation excluded)
- Any risk of haemorrhage



Dosage and Administration

Patients will receive Enoxaparin co-therapy as an intravenous bolus of 30 mg followed by the first subcutaneous dose of 1 mg/kg about 15 minutes after the administration of tenecteplase

Packaging

- Clexane 40: Enoxaparin 40 mg per 0,4ml
- Clexane 60: Enoxaparin 60 mg per 0,6ml
- Clexane 80: Enoxaparin 80 mg per 0,8ml
- Clexane 100: Enoxaparin 100 mg per 1,0ml
- Clexane 300: Per multidose vial. Enoxaparin sodium 300 mg, benzyl alcohol (preservative) 1, 5% m/v, Water for injection to 3, 0 ml

Heparin Sodium

Anticoagulant



Pharmacological Action

Acts at multiple sites in the normal coagulation system. Interacts with anti-thrombin to change its configuration and enhance its ability to inhibit thrombosis by inactivating clotting factor proteases, especially thrombin (IIa), Ixa and Xa by forming equimolar complexes with them. Inactivates thrombin and factor Xa through anti-thrombin dependent mechanisms. Inactivation of thrombin prevents fibrin formation and inhibits thrombin-based activation of platelets.

Indications

- STEMI
- Non-ST elevation ACS in patient with cardiogenic shock

Adverse Effects

- Haemorrhage
- Thrombocytopenia
- Venothrombosis (PE/ DVT)

Packaging

Heparin/ Bio-heparin: 25 000 IU in 5ml vials

Contra-indications

- Uncontrolled haemorrhage
- Thrombocytopenia
- Hypertension
- Hepatic/ kidney disease
- Known allergy
- Not to be administered if patient is receiving thrombolytic therapy

Onset of action: Immediate
Duration of action: 3-6 hours

Dosage and Administration

<u>Adult</u>

STEMI: 5000 IU IV bolus or 333IU/kg subcutaneous injection → Followed by 1000-1200 IU/ hour IV infusion

Hydralazine

Anti-hypertensive/ Vasodilator

Pharmacological Action

Hydralazine acts directly on arteriolar smooth muscle causing relaxation and subsequent arteriolar dilation. Hydralazine does not relax venous smooth muscle. Arteriolar dilation causes decrease in BP. Hydralazine has stimulatory effect on sympathetic system, which can result in tachycardia and tachyphylaxis.

Indications

Recommended for pre-hospital treatment of pre-eclampsia/ eclampsia for control of hypertension

Contra-indications

- Allergy
- Coronary artery disease
- Mitral valve rheumatic heart disease

Onset and duration is dose dependent

Adverse Effects

- CNS → Dizziness, drowsiness, headache
- CVS → Tachycardia, arrhythmias, angina, orthostatic hypotension
- Edema
- GIT → Diarrhea, vomiting, nausea
- CNS → Peripheral neuropathy

Packaging

- Tablets: 10mg, 25mg, 50mg, 100mg
- Injection: 20mg/ml



Dosage and Administration

<u>Adults</u>

- Oral: 10mg four times a day
- IM/ IV: 5-40mg repeated as needed
- Eclampsia: 5mg every 15-20 mins; if no response after total of 20mg, consider alternative agent.

Labetalol

Anti-hypertensive/ Beta-blocker

Pharmacological Action

Non-selective beta-receptor blocker with intrinsic sympathomimetic activity, which decreases peripheral resistance by blocking peripheral arteriolar alpha-adrenoreceptors while causing a concurrent beta-blockade - thus protecting the heart from reflex sympathetic drive and maintaining adequate cardiac output .

Dosage and Administration

Adults

Severe Hypotension:

- 50mg over 1 min (every 5 min); Max dose 200mg
- IV infusion → 1mg/ml solution @ 2-8mg/min (titrate to supine BP)
- Oral → 200mg every 12 hours; Max dose 400mg/ 24 hours

Hypertension in pregnancy:

- IV infusion → 1mg/ml solution @ 20mg/hour – titrate by doubling dose every 30 mins until desired effects; max dose 160mg/ hour

Onset of action: 2-5 min IV; 20-120 min Oral
Duration of action: 2-3 hours

Indications

- Hypertensive emergencies
- Suspected MI/ UA
- Adjunctive therapy with fibrinolysis
- SVT to convert to sinus rhythm or slow the ventricular response

Contra-indications

- Allergy
- Severe hepatic impairment
- Cardiogenic shock/ Heart failure/ Heart blocks/ Sick sinus syndrome
- Asthma/ COPD
- Metabolic acidosis
- Severe bradycardia
- Hypotension

Adverse Effects

- Light headedness, fatigue
- Nausea
- Hypotension
- Bradycardia
- Sweating
- Bronchospasm (rare)

Packaging

- Injectable solution: 5mg/ml
- Tablets: 100mg, 200mg, 300mg



Nefidipine

Anti-hypertensive/ Calcium channel blocker

Pharmacological Action

Inhibits influx of calcium ions through L-type calcium channels, leading to: Inhibition of contractile smooth muscle cells; dilation of coronary and systemic arteries; increased O2 delivery to the myocardium; decreased total peripheral resistance; decreased systemic BP; Decreased afterload

Indications

Pre-hospital: Preterm labour gestation < 34 weeks (Tocolysis)

Packaging

10mg and 20mg tablets

Contra-indications

- Hypotension
- Cardiac disease (CCF, aortic stenosis)
- Hepatic dysfunction
- MI < 4 weeks



Onset of action: 10-30 mins
Duration of action: 4 hours

Adverse Effects

- Flushing
- Headache
- Constipation
- Heart block
- Worsens heart failure

Dosage and Administration

- Tocolysis: Give oral stat dose of 20mg → repeat after 30min if uterine contractions persist
- Hypertensive emergency in pregnancy: Oral 10mg stat → Repeat after 30 min if BP > 160/110

Precautions

- Concomitant use of MgSO4 may result in significant hypotension and neuromuscular weakness

Furosemide

Diuretic



Pharmacological Action

Furosemide is a loop diuretic acting primarily by inhibiting electrolyte (Na+) and fluid re-absorption in the ascending limb of the loop of Henle → increased excretion of sodium, chloride, potassium, calcium and magnesium – water follows passively. In patients with pulmonary edema, furosemide increases systemic venous capacitance, thereby decreasing left ventricular filling pressure (preload).

Indications

Acute pulmonary edema of cardiac, hepatic or renal origin

Adverse Effects

- Hyponatremia, hypokalemia
- Hypotension
- Hypovolemia
- Tinnitus and deafness (following rapid IV administration)

Contra-indications

- Known allergy
- SBP < 90mmHg
- Hypovolemia/ dehydration
- Hypokalemia

Precautions

- Urinary obstruction or retention
- Elderly patients are particularly susceptible to dehydration and hypotension

Dosage and Administration

- Adults in CHF: 0.5-1mg/kg slowly over 2 min (max dose per min – 20mg); loading dose may be repeated every 1-2 hours → 4-60mg/hr as an ongoing infusion after loading dose
- Acute pulmonary edema: 40-80mg IV initial dose → double the dose after 4-6 hours if needed

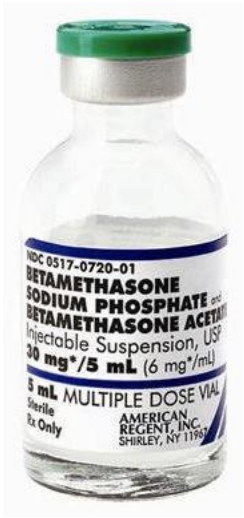
Onset of action: 5 min
Duration of action: 120 min

Packaging

- 20mg in 2ml ampoule
- 50mg in 5ml ampoule
- 250mg in 25ml ampoule

Betamethasone

Glucocorticosteroid/ Anti-inflammatory



Pharmacological Action

Prostaglandins and leukotrienes are inhibited by glucocorticoids. Glucocorticoids stimulate the production of lipocortin-1, which acts in the extracellular space, where it binds to the leukocyte membrane receptors and inhibits various inflammatory events. Additionally, the immune system is suppressed due to a decrease in lymphatic function, reduction in immunoglobulin and complement concentrations, precipitation of lymphocytopenia, and interference with antigen-antibody binding.

Packaging

- 3mg/ml vial
- 4mg/ml vial
- 6mg/ml vial

Indications

- Anti-inflammatory
- Immune-modulation

Contra-indications

- Allergy
- Systemic infections
- Injection into tendons

Adverse Effects

- CNS → Psychotic reactions, irritability/ anxiety,confusion, seizures, increased ICP
- CVS → Hypertension, MI, Thromboembolsim
- GIT → Abdominal distention, ulceration, nausea
- Metabolic → Hypokalemia, sodium and water retention

Dosage and Administration

Adults

4-20mg IM repeated 6 hourly

Dexamthasone

Glucocorticosteroid/ Anti-inflammatory



Pharmacological Action

Dexamethasone is a glucocorticoid agonist with properties that adjust sodium retention and blocks leukocyte infiltration for the interference with the inflammatory mediators.

Indications

- Preterm labour
- Asthma and anaphylaxis to reduce secondary effect

Packaging

- 4mg/ml vial
- 6mg/ml vial
- 10mg/ml vial
- 2,4 and 6mg tablets

Adverse Effects

- Blurred vision
- Headache
- Anxiety
- SOB
- Fluid retention

Contra-indications

- Allergy
- Liver impairment
- Myasthenia gravis
- CCF
- Hypertension
- Depression/ mental illness
- Thyroid disease
- Malaria and TB

Onset of action: 10-30 min
Duration of action: 2-4 days

Dosage and Administration

Adults

- Anaphylaxis/ Asthma: 4-8mg IM single dose
- Preterm labour: 6mg every 12 hours over 24 hours

Oxytocin

Uterotonic

Pharmacological Action

Oxytocin is produced by the pituitary gland. Pitocin is a man-made version of oxytocin used for stimulating contraction of the uterus. Oxytocin works by increasing the concentration of calcium inside muscle cells that control contraction of the uterus.

Contra-indications

- Allergy
- Significant cephalopelvic disproportion
- Unfavorable foetal positions
- Foetal distress when delivery is not imminent
- Hyperactive or hypertonic uterus
- Ecclampsia; uterine atony

Indications

- Labour induction
- Active management of 3rd stage of labour
- Management of PPH

Dosage and Administration

- Initial dose: 10IU IMI
- Alternative: Infusion of 20-40 IU in 1000ml @ 150ml/hour
- Max dose: 40IU to 1000ml

Onset of action: Immediate (IV)
Duration of action: 1 hour (IV)

Adverse Effects

- Nausea & vomiting
- Bleeding
- Hypertension
- Fetal bradycardia
- PVCs
- Blurred vision
- Hyperbilirubinemia
- Arrhythmia exacerbation
- Pulmonary edema
- Seizures

Packaging

- 5IU/ ml vial
- 10IU/ ml vial



Tranexamic Acid

Antifibrinolytic

Pharmacological Action

TXA competitively inhibits the activation of plasminogen to plasmin by forming a reversible complex that displaces plasminogen from fibrin resulting in inhibition of fibrinolysis. It also inhibits the proteolytic activity of plasmin.

Contra-indications

- Known allergy
- Active thromboembolic disease (DVT/ PE)
- History of thrombosis/ thromboembolism
- Severe renal impairment
- History of seizures

Indications

- Traumatic haemorrhage and bleeding
- Post-partum haemorrhage

Onset of action: 1 hour
Half-life: 3 hours

Adverse Effects

- CNS → Headache, fatigue, dizziness, seizures
- CVS → Hypotension (with rapid IV administration), retinal artery and vein occlusion
- GIT → Nausea, vomiting, diarrhea, abdominal pain
- Anaphylaxis
- Musculoskeletal pain
- Visual disturbances

Dosage and Administration

- Trauma-associated haemorrhage – 1g over 10 mins loading dose, followed by 1g over 8 hours (use within 3 hours of injury)
- PPH – 1g over 10 mins (within 3 hours of birth). Followed by another 1g over 10 mins if bleeding continues after 30 mins of 1st dose.



Packaging

- IV: 100mg/ml ampoule
- Oral: 500mg capsule

Hydrocortisone

Corticosteroid

Pharmacological Action

Corticosteroids are anti-inflammatory, anti-pruritic and vasoconstrictive. Hydrocortisone has a catabolic mechanism of action which consists of glucocorticoid properties. It has weak anti-inflammatory effects, but acts as a powerful mineralocorticoid. Hydrocortisone causes:

- Suppression of (1) inflammatory/ allergic responses by inhibiting polymorphonuclear leukocyte production in the area of inflammation; (2) immunological responses (antibody production)
- Stabilization of mast cell membranes
- Upregulation of β2 receptor responsiveness in asthma.

Adverse Effects

- Hyperglycemia
- Neuropsychiatric reactions
- GIT disturbances
- Fluid and electrolyte disturbances
- Hypertension
- Susceptibility to infections and leukocytosis



Packaging

100mg/2ml vial
500mg/4ml vial

Onset of action: 1-2 hours
Duration of action: 6-12 hours

Dosage and Administration

Adults	Children
• Anaphylaxis – 200-500mg IM or IV • Moderate-severe asthma – 5mg/kg IV slowly	• Anaphylaxis – 50-100mg IV or IM • Moderate-severe asthma – 5mg/kg IV slowly

Precautions

- Not to be used in cerebral edema associated with TBI/ stroke
- Emotional instability/ psychiatric disorders
- Epilepsy
- Thromboembolic disorders
- Peptic ulcer (extreme caution)
- Hypertension and CCF
- Diabetes
- Liver cirrhosis
- Osteoporosis

Indications

- Severe allergy/ Anaphylaxis
- Acute asthma attack

Methylprednisilone

Corticosteroid

Pharmacological Action

Mthylprednisolone inhibits leukocyte infiltration at the inflammation site. It interferes with inflammatory mediators and suppresses humoral immune response. The anti-inflammatory actions are considered to involve phospholipase A2 inhibitory proteins and lipocortins which control the biosynthesis of potent inflammatory mediators such as prostaglandins and leukotrienes.

Adverse Effects

- Hyperglycemia
- Neuropsychiatric reactions
- GIT disturbances
- Fluid and electrolyte disturbances
- Hypertension
- Susceptibility to infections and leukocytosis



Indications

- Severe allergy/ Anaphylaxis
- Acute asthma attack

Onset of action: 1-2 hours
Duration of action: 30-36 hours

Precautions

- Liver cirrhosis
- Hypertension
- Diverticulitis
- Peptic ulcer
- Osteoporosis
- Psychiatric disorders
- Diabetes
- Epilepsy
- Thromboembolic disorders
- Pregnancy
- Geriatrics

Packaging

- 40mg/ml vial
- 125mg/2ml vial
- 500mg/8ml vial
- 1000mg/16ml vial

Dosage and Administration

Adults	Children
Initial bolus: 125mg IV slowly Range: 40-125mg IV	Initial bolus: 1mg/kg IV slowly Max dose: 30mg



Prednisilone

Glucocorticosteroid

Pharmacological Action

Crosses cell membranes and binds to cytoplasmic receptors – Inhibits leukocyte infiltration at inflammation site which causes interference of the inflammatory mediators' response, suppression of the humoral immune system responses, including reduction of edema as well as scar tissue.

Dosage and Administration

Adults	Children
0.5-1mg/kg	0.5-1mg/kg
• Anti-inflammatory: 5-60mg • Acute asthma: 40-80mg	• Anti-inflammatory: 5-60mg • Acute asthma: 1-2mg/kg

Indications

- Severe allergy/ Anaphylaxis
- Acute asthma attack

Onset of action: 1 hour
Duration of action: 24 hours

Adverse Effects

- CNS → Anxiety; psychotic disorders; irritability; epilepsy aggravation; cognitive dysfunction
- CVS → Hypertension; edema; thromboembolism; cardiac failure
- GIT → Abdominal distension; nausea; dyspepsia; oesophageal ulceration; peptic ulceration
- Endocrine → Hyperglycemia; hypokalemia; sodium and water retention

Packaging

- Lenisolone: 5mg tablets
- Adco-prednisolone: 15mg/5ml
- Prelone: 15mg/ml; 5mg tablets

Precautions

- Pregnancy (Safety not established)
- Lactation (Safety not established)
- Renal impairment
- Elderly
- Diabetes mellitus
- Cardiovascular disease
- Hypertension
- Epilepsy
- Peptic ulcers
- Hepatic impairment
- Hypothyroidism
- Recent MI
- Acute psychosis
- Tuberculosis

Ipratropium Bromide

Bronchodilator/ Anticholinergic



Pharmacological Action

Ipratropium bromide inhibits the contraction of airway smooth muscles and mucus secretion (bronchoconstriction) that is vagally mediated, enabling bronchodilation. Antagonizes the action of acetylcholine on airway smooth muscle cells.

Dosage and Administration

Adults	Children
• Aerosol: 1 puff (40ug) 3-4 times a day • Nebulized solution: 0.5mg diluted with 3-5ml NS	• Aerosol: 1 puff (40ug) 3-4 times a day • Nebulized solution: 0.25 mg diluted with 3-5ml NS

Adverse Effects

- Xerostomia
- Bitter taste in mouth
- Rare disorders: Constipation; Paradoxical bronchospasm; Urinary retention

Packaging

- Unit dose vials: 0.25mg/2ml or 5mg/2ml
- Nebulizer solution: 0.25mg/ml
- Metered dose inhaler: 40ug/dose

Precautions

- Benign prostatic hypertrophy: may cause urinary retention
- May aggravate narrow-angle glaucoma
- Myasthenia gravis: use with caution
- Hypersensitivity reaction: discontinue use and administer alternative treatment
- Not recommended as rescue therapy, only use for maintenance treatment
- Bronchospasm: inhaled bronchodilators should be differentiated from insufficient response as they may cause life threatening paradoxical bronchospasm.
- Pregnancy
- Breastfeeding

Ipratropium bromide + β2 stimulant = synergistic effect. May be particularly useful in pt who have taken β-blockers. Typically only administered once due to prolonged onset of action.

Indications

- COPD exacerbation
- Asthma exacerbation

Onset of action: 30 min
Duration of action: 4-6 hours

Salbutamol/ Fenoterol

B2 Stimulants

Indications

- Acute bronchospasm
- Premature/ obstructive labour
- Suspected hyperkalemia on ECG

Pharmacological Action

Selective β2 stimulants acting on β2 receptors in the lungs → bronchodilation. At higher/ repeated doses, the systemic absorption progressively increases, thus acting on other organs with β2 receptors:

- Skeletal muscle contraction
- Vasodilation
- Tocolysis (uterine smooth muscle relaxation)
- Break down of glycogen to glucose

At higher/ repeated doses, the selectivity is also progressively lost and β1 effects (myocardium) are experienced:

- Positive inotrope, positive chronotrope; positive dromotrope
- Increased myocardial oxygen consumption

Dosage and Administration

Acute Bronchospasm

- Feneterol: Adults – 1.25/2ml + 3ml NS nebulized; Children – 0.5mg/2ml + 3ml NS nebulized
- Salbutamol: 5mg/ml + 4ml NS nebulized (halved for paediatrics)

Tocolysis

Salbutamol

- Bolus: 100-250ug IV slowly → Dilute 500ug/ml ampoule with 19mlNS = 500ug/20ml = 25ug/ml → Administer 1ml/min until (1) total of 250ug/ 10 min given; (2) Maternal HR >120bpm; (3) Contractions cease.
- Maintenance infusion: Dilute 2mg into 200ml NS = 10ug/ml → Start with 10ug/min – may increase by 10ug/min every 10 min to max dose of 45ug/min.



Packaging

Fenoterol

- Aerosol: 100ug
- Inhalant solution: 1mg/ml
- UDV: 1.25mg/2ml or 0.5mg/2ml

Salbutamol

- Ventolin aerosol: 100ug
- Resp. solution: 5mg/ml
- UDV: 2.5 or 5mg/2.5ml
- IV solution: 0.5-1mg/ml

Adverse Effects

- Tremors, restlessness, anxiety, confusion, headache
- Hypotension, tachycardia, palpitations
- Cramps, nausea, vomiting
- Urinary retention
- Hyperglycemia
- Hypokalemia

Precautions

- HR > 120bpm
- IV β2 should be used with caution in:
- Ischemic heart disease, cardiac arrhythmias, cardiac failure
- Occlusive vascular disorders, hyper-/ hypotension
- Hyperthyroidism; diabetes

Onset of action: 5-15 min
Duration of action: 3-6 hours

Fentanyl

Synthetic narcotic agonist/ Opioid

Onset of action: 1-2 min (IV); 7-15 min (IM)
Duration of action: 30-60 min (IV); 60-120 min (IM)

Pharmacological Action

Fentanyl, like other opioids act mainly by activating the μ (mu)-opioid peptide (MOP) receptors, while also stimulating some activity on the kappa (k) receptors. Respiratory depression, analgesia, sedation and euphoria are effects that result from activation of μ (mu)-opioid peptide (MOP) receptor. Innervation of the kappa receptors only leads to analgesia and sedation.

Precautions

- Fentanyl Is 100 times more potent than morphine.
- In pregnancy, rapid Intravenous fentanyl administration may result in muscle rigidity.
- May cause respiratory depression in neonates of mothers who receive this drug in labour.
- It may increase risk of overdose in patients using phenothiazines, ethanol, CNS depressants and tricyclic antidepressants.
- Herbal interactions effect include increased Central Nervous System depression with the use of kava kava, St John's wort, and gotu kola.

Indications

- Procedural sedation (PSA)
- Pre-treatment
- Induction
- Analgesia



Packaging

10ug/1ml ampoule
100, 200, 400, 600, 800ug tablets

Dosage and Administration

2-20ug/kg depending on indication
Intra-nasal administration for adults & children: 1.5ug/kg

Contra-indications

- Known allergy
- GCS < 15
- Hypotension
- Hypoxia (SpO2 < 90%) despite O2 delivery
- Signs of hypoventilation

Adverse Effects

- Resp → Bronchoconstriction; depression
- CVS → Hypotension, flushing, sweating, bradycardia
- CNS → Depression, convulsions
- GIT → Dry mouth, biliary spasm, nausea, vomiting
- Other → Miosis, bluured vision; urinary retension; histamine release

Morphine Sulphate

Narcotic analgesic

Pharmacological Action

Morphine is a centrally acting analgesic that binds to specific opioid receptors in the brain and spinal cord, resulting in an increase of the pain threshold. It reduces myocardial oxygen consumption and workload. Morphine dilates the capacitance vessels of the peripheral venous bed, reducing venous return and diminishing ventricular preload. Morphine also reduces afterload by causing mild arterial vasodilation, making it an acceptable adjunct for acute pulmonary edema.

Precautions

- CNS disorders → Head injury, raised ICP
- CVS disorders → Hypotension, hypovolemia
- Resp disorder → COPD, asthma
- GIT disorders → Undiagnosed abdominal pain
- Children < 1 year old/ Elderly
- Effects of opiates may be enhanced by alcohol, barbiturates, benzodiazepines, narcotics and other CNS depressants



Adverse Effects

- Resp → Bronchoconstriction; depression
- CVS → Hypotension, flushing, sweating, bradycardia
- CNS → Depression, convulsions
- GIT → Dry mouth, biliary spasm, nausea, vomiting
- Other → Miosis, bluured vision; urinary retension; histamine release

Indications

- Acute severe pain
- ACS not relieved by GTN
- Cardiogenic pulmonary edema
- Concominant use with benzodiazepines for synergism in induction

Packaging

- 10mg/ml amber coloured ampoule
- 15mg/ml amber coloured ampoule

Dosage and Administration

0.1mg/kg

Dilute to concentration of 1mg/ml and titrate to effect administering very slowly

Onset of action: 2-3 min with peak effect @ 20 min

Duration of action: 4-6 hours

Nitrous Oxide

Analgesic gas

Pharmacological Action

Entenox is a colourless, sweet-smelling gas that is heavier than room air/ oxygen. Nitrous oxide has mild analgesic and anaesthetic effects depending on the dose inhaled by depressing the CNS. The high concentration of O2 delivered with the NO increases oxygen tension in the blood, thereby reducing hypoxia.

Precautions

- The constituent gases dissociate at <4°C; cylinder must be inverted a few times then placed horizontal when used otherwise pt will inhale pure nitrous oxide.
- When the mask is removed after prolonged use, the gas will come out of solution iin the lungs and displace O2 in the alveoli. Causing hypoxia → pt must receive oxygen for 5-10 min following prolonged use.



Contra-indications

- Neurological impairment → Any altered LOC; Head injury
- Air entrapment → COPD/ asthma; acute pulmonary edema; chest injury; abdominal trauma; diving accidents; burns to the respiratory tract
- Other → Hypotension (SBP<90mmHg); major facial trauma

Adverse Effects

- Light-headedness
- Drowsiness
- Nausea and vomiting

Packaging

Pressurized cylinder containing 52% nitrous oxide and 48% oxygen

Onset of action: 30-60 sec

Indications

Relief of pain from:

- AMI
- Musculoskeletal trauma
- Burns (excluding respiratory tract)
- Active labor

Dosage and Administration

Sefl-administered gas: explain procedure to patient and monitor for effects; if pt becomes drowsy, remove entenox and replace with oxygen.

Paracetamol

Analgesic/ Antipyretic

Pharmacological Action

Increases pain threshold and readjusts hypothalamic temperature regulatory center by inhibiting both isoforms of cyclooxygenase, COX -1, COX-2, and COX-3 enzymes involved in prostaglandin synthesis.

Adverse Effects

- CVS → hypotension, tachycardia
- Hepatobiliary → Increased hepatic transaminases, hepatic dysfunction
- Subcutaneous tissue → Allergic reactions, erythema, flushing, pruritis

Onset of action: IV 5-10min (analgesic); 30 min (antipyretic)
Duration of action: 4-6 hours

Packaging

- Oral: 500mg tablet or capsule
- IV Infusion: 1g/100ml

Precautions

- Pregnancy
- Renal impairment
- Mild to moderate hepatic impairment
- Risk of serious liver damage with higher than recommended dose
- Chronic alcoholism
- Chronic malnutrition
- Dehydration

Dosage and Administration

<u>Adult</u>

- Oral: 500-1000mg 4-6 hourly
- IV: 1000mg 4-6 hourly

<u>Children</u>

- Oral: 10-15mg/kg/dose 4-6 hourly
- IV: 15mg/kg 6 hourly

Indications

Mild to moderate pain and fever

Contra-indications

Allergy; mild to moderate renal or hepatic impairment



Penthroxyflurane

Analgesic

Dosage and Administration

Up to 6 ml (2 x 3 ml bottles) of PENTHROP per day, vaporised in a PENTHROP inhale – self administered by patient.

Pharmacological Action

Methoxyflurane vapour possesses analgesic properties when inhaled at low concentrations. The precise mechanism of action whereby methoxyflurane produces analgesia at sub-anaesthetic doses is unknown, although a reduction in substance P- and β-endorphine-like immunoreactivity in the brain has been suggested. After methoxyflurane administration, drowsiness may occur. The myocardium is moderately sensitised to epinephrine (adrenaline) by methoxyflurane. The blood pressure decreases in a dose dependant manner. This may be accompanied by bradycardia. The blood pressure decrease noted is accompanied by reduced cardiac contractile force and reduced cardiac output.

Contra-indications

- Renal impairment, including reduced glomerular filtration rate (GFR), urine output and reduced renal blood flow
- Severe hepatic impairment
- Cardiovascular instability
- Respiratory depression
- Head injury or loss of consciousness
- Malignant hyperthermia: patients with known or genetically susceptible to malignant hyperthermia
- Porphyria
- Hypersensitivity

Adverse Effects

- CVS → Hypotension, bradycardia
- CNS → Retrograde amnesia, drowsiness, dizziness, headache
- Resp → Respiratory depression, bronchospasm, laryngospasm, coughing
- Nausea, vomiting, polyuria
- Fever, malignant hyperthermia



Indications

For emergency relief of pain by self-administration in conscious haemodynamically stable patients with trauma and associated pain

Packaging

Combination pack containing one sealed bottle filled with 3 ml liquid, one PENTHROP inhaler and one Activated Carbon Scavenger in an outer carton box (pack of 1 or 10 units).

Diazepam

Benzodiazepine/ Anticonvulsant/ Anxiolytic

Packaging

10mg/2ml amber coloured ampoule

Indications

- Seizures
- Sedation

Onset of action: 1-5 mins
Duration of action: 15-120 mins

Pharmacological Action

Benzodiazepines are GABA agonists. Binding of GABA to its receptor triggers an opening of the central ion channel, allowing chloride through the pore. The influx of chloride ions causes hyperpolarization of the neuron and decreases neurotransmission by inhibiting the formation of action potentials. Benzodiazepines increase the frequency of channel openings produced by GABA. Binding of a benzodiazepine to its receptor site increases the affinity of GABA for the GABA-binding site (and vice versa).

Precautions

- CVS → CCF, Hypotension, Hypovolaemia
- Respiratory Disorders → Asthma, COPD, Hypoventilation
- Active Labour → Neonatal Suppression
- Psychosis → Possible increase in agitation, Offers no anti-psychotic effects
- Absorption of Diazepam from intramuscular injections remains unreliable and erratic.
- Reversible causes of convulsions, for example hypoglycaemia, should be ruled out before administration of Diazepam.
- The effects of Diazepam may be enhanced or altered by other narcotics and depressants of the central nervous system, including barbiturates, and alcohol.

Dosage and Administration

<u>Adults</u>

- Convulsions: 5mg/min slowly IVI (0.15mg/kg) every 2-5 min – Titrate to effect; Max dose: 20mg
- Sedation: 1mg every 30sec IV titrated to effect

<u>Children</u>

- Convulsions: 0.2mg/kg slowly IVI every 2-5 min – Titrate to effect; Max dose: children > 5 years = 10mg; children < 5 years = 5mg

Contra-indications

- Allergy to benzodiazepines
- An inability to ventilate patient, should the need arise (due to side effect of respiratory control depression)

Adverse Effects

- Central nervous system depression
- Depression of respiratory control (especially following rapid administration)
- Acute hypotension (especially following large dosage or rapid administration)
- Diplopia
- Paradoxical excitation
- Nausea and vomiting
- Thrombophlebitis



Lorazepam

Benzodiazepine/ Anticonvulsant/ anxiolytic

Packaging

4mg/ml ampoule

Indications

- Seizures – first-line

Onset of action: 3 mins
Duration of action: 12-24 hours

Pharmacological Action

Benzodiazepines are GABA agonists. Binding of GABA to its receptor triggers an opening of the central ion channel, allowing chloride through the pore. The influx of chloride ions causes hyperpolarization of the neuron and decreases neurotransmission by inhibiting the formation of action potentials. Benzodiazepines increase the frequency of channel openings produced by GABA. Binding of a benzodiazepine to its receptor site increases the affinity of GABA for the GABA-binding site (and vice versa). Lorazepam has longer CNS activity than Diazepam making it the preferable first-line anticonvulsant.

Precautions

- CVS → CCF, Hypotension, Hypovolaemia
- Respiratory Disorders → Asthma, COPD, Hypoventilation
- Active Labour → Neonatal Suppression
- Psychosis → Possible increase in agitation, Offers no anti-psychotic effects
- Absorption of Diazepam from intramuscular injections remains unreliable and erratic.
- Reversible causes of convulsions, for example hypoglycaemia, should be ruled out before administration of Diazepam.
- The effects of Diazepam may be enhanced or altered by other narcotics and depressants of the central nervous system, including barbiturates, and alcohol.



Dosage and Administration

<u>Adults</u>

2-4mg slowly IVI; if needed repeat 4mg dose in 10 min → Titrate to effect; Max dose: 8mg/12 hours

<u>Children</u>

0.05-0.1mg/kg slowly IVI up to max single dose of 4mg; repeat after 10min → Titrate to effect

Contra-indications

- Allergy to benzodiazepines
- An inability to ventilate patient, should the need arise (due to side effect of respiratory control depression)

Adverse Effects

- Central nervous system depression
- Depression of respiratory control (especially following rapid administration)
- Acute hypotension (especially following large dosage or rapid administration)
- Diplopia
- Paradoxical excitation
- Nausea and vomiting
- Thrombophlebitis

Midazolam

Benzodiazepine/ Anticonvulsant/ anxiolytic

Precautions

Same as for other Benzodiazepines

Onset of action: 1-3 mins
Duration of action: 30 mins

Pharmacological Action

Benzodiazepines are GABA agonists. Binding of GABA to its receptor triggers an opening of the central ion channel, allowing chloride through the pore. The influx of chloride ions causes hyperpolarization of the neuron and decreases neurotransmission by inhibiting the formation of action potentials. Benzodiazepines increase the frequency of channel openings produced by GABA. Binding of a benzodiazepine to its receptor site increases the affinity of GABA for the GABA-binding site (and vice versa). Midazolam is water-soluble, making it suitable for mixture and infusion, unlike oil-based Diazepam.

Dosage and Administration

<u>Adults</u>

- Sedation: 1mg/min slowly IVI
- Induction: 1mg/ 15-30 sec (Titrate to effect); Maintenance infusion: 0.03-0.1 mg/kg/hour in combination with narcotic analgesics
- Seizures: 0.15mg/kg slowly IVI
- IM only if no IV access: 0.15mg/kg

<u>Children</u>

- Sedation: 0.5mg/min slowly IVI
- Induction: 0.5mg/ 30 sec IVI (Titrate to effect); Maintenance infusion: Same as for adult
- Seizures: 0.15mg/kg slowly IVI/ IMI; Rectally: 0.4-1mg/kg; Nasally: 0.4mg/kg



Packaging

5mg/5ml ampoule
15mg/3ml ampoule
50mg/10ml ampoule

Indications

- Sedation
- Induction
- Seizures

Contra-indications

- Allergy to benzodiazepines
- An inability to ventilate patient, should the need arise (due to side effect of respiratory control depression)

Adverse Effects

- Central nervous system depression
- Depression of respiratory control (especially following rapid administration)
- Acute hypotension (especially following large dosage or rapid administration)
- Diplopia
- Paradoxical excitation
- Nausea and vomiting
- Thrombophlebitis

Etomidate

Sedative/ Hypnotic

Onset of action: 1 min
Half-life: 3-5 hours

Pharmacological Action

Etomidate increases the sensitivity of the γ-aminobutyric acid (GABAa) receptors to the inhibitory neurotransmitter GABA. This increases chloride ion influx and hyperpolarization of neurons. Postsynaptic neuronal excitability and, thus, CNS activity are diminished. When used for induction of anaesthesia, Etomidate causes adrenal suppression by producing lowered plasma cortisol levels and aldosterone

Precautions

In haemodynamically unstable patients, use a lower dosage of Etomidate. Beware of over dosage in obese patients. A maximum body weight of 100kg should be used for calculating dosages. Other sedatives used in conjunction with Etomidate may potentiate it's effect.

Adverse Effects

- CNS → Dyskinesia; myoclonus; nystagmus
- CVS → hypotension; bradycardia
- Resp → Apnoea; hyperventilation; stridor
- GIT → Nausea, vomiting

Contra-indications

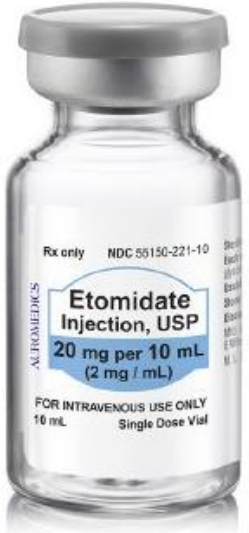
Etomidate is contra-indicated in patients with a known hypersensitivity to Etomidate. It should not be used in patients with adrenocortical dysfunction as it suppresses adrenal functioning.

Dosage and Administration

- Adults: 0.1 – 0.3 mg/ kg IV
- Children <15 years: May require a higher dose of 30% more → 0.1 – 0.4 mg/kg.
- Elderly >65 years: 0.1 – 0.2mg/kg titrate to effect

Indications

As an induction agent for RSI (Rapid Sequence Intubation) and procedural sedation in adults and children.



Packaging

20mg/10ml ampoule

Ketamine

Dissociative Anaesthetic

Packaging

10mg/ml; 50mg/ml;
100mg/ml multi-dose vials

Adverse Effects

- CNS → Emergence syndrome (hallucinations)
- CVS → Hypertension; tachycardia
- GIT & Resp → increased secretions



Dosage and Administration

<u>Adults</u>

- Induction: 1-4.5mg/kg slowly IVI (over 30-60 sec)
- Maintenance: IV infusion 1-5mg/kg/hour
- Analgesia & Sedation: 0.5-2mg/kg IM/IVI → Repeat as necessary

<u>Children</u>

- Induction: 1.5-2mg/kg slowly IVI (over 30-60 sec)
- Maintenance: IV infusion of 11mg/kg/hour from 0-20 mins → 7mg/kg/hour from 20-40min → 5mg/kg/hour from 40-60min → 4mg/kg/hour from 1-2 hours

Pharmacological Action

Ketamine is lipophilic, which allows it to cross the blood brain barrier and enter the central nervous system (CNS) quickly. Ketamine is believed to act by blocking the activation of N-methyl-D-aspartate (NMDA) receptors at the GABA-complex. This results in inhibition of the CNS and anaesthesia. Ketamine also act on the opioid receptors producing profound analgesia. It produces a state known as “dissociative anaesthesia”, characterised by sensory loss, analgesia and amnesia, without complete loss of consciousness. Ketamine has a different effect on the cardiovascular system (CVS), when compared to other anaesthetics. It stimulates central sympathetic outflow resulting in increased blood pressure, cardiac output and heart rate. Therefore this drug is contra-indicated in patients with cardiac pathologies. Ketamine is a powerful bronchodilator and respiration is mostly unaffected by effective anaesthetic doses. Ketamine is thus especially useful in hypovolemic patients and asthmatics. Consider 0.03 mg/kg prophylactic midazolam in both adults and children, in order to decrease the risk of unpleasant emergence reactions.

Contra-indications

- Hypertension
- Aortic dissection
- Cardiac failure
- Recent myocardial infarction
- Ischemic heart disease
- Acute psychiatric disorders
- Thyrotoxicosis

Indications

- Induction agent for rapid sequence intubation including patients with TBI or increased ICP
- Recommended sedative in asthmatic patients
- Procedural sedation and analgesia in adults and children

Onset of action: 30-60 sec
Duration of action: 5-10 mins

Ondansetron

Anti-emetic



Pharmacological Action

Ondansetron is a selective 5-HT3 receptor antagonist The effect of ondansetron in the management of nausea and vomiting, may be due to the antagonism of 5-HT3 receptors on neurons located in the peripheral and central nervous system. Thus, the activation of the vomiting reflex is inhibited. Ondansetron reduces the activity of the Vagus nerve, which activates the vomiting center in the medulla oblongata, and blocks serotonin receptors in the chemoreceptor trigger zone.

Dosage and Administration

Adults: Nausea and/or Vomiting

- 4 – 8 mg, total max dosage 8 mg (PO, IM, IV)

Paediatrics:

- ≥ 5 years – 4 mg, PO (single dose only)
- ≥ 5 years – 2 mg, IM (single dose only)
- ≥ 3 years – 100 mcg/kg, IM (single dose only; don't exceed 4 mg)
- ≥ 3 years – 100 mcg/kg, slowly over 2 -3 mins IV (single dose only; don't exceed 4 mg)

Contra-indications

- Known hypersensitivity to Ondansetron
- Known Long-QT syndrome
- Congenital QT prolongation (may lengthen QT interval) – patients should be placed on cardiac monitoring after administration
- Current apomorphine (dopamine agonist used in the treatment of Parkinson's disease) therapy
- Paediatrics < 3 years

Adverse Effects

- CNS → Headache; extrapyramidal reactions; dyskinesia; dizziness; depression
- CVS → Chest pain; arrhythmias; bradycardia; QT prolongation/ Torsades de pointes; flushing; hypotention
- GIT → Constipation
- Anaphylaxis; transient visual disturbances

Indications

Severe, persistent nausea and/or vomiting

Onset of action: 30 mins
Duration of action: 2-7 hours

Packaging

4mg/2ml ampoule

Metoclopramide

Anti-emetic

Pharmacological Action

Metoclopramide acts on the chemo-emetic trigger zone (CETZ) to produce a central anti-emetic effect. It also enhances he motility of GIT smooth muscle from the esophagus through the proximal small bowel and accelerates gastric emptying and the transit of intestinal contents from the duodenum to the ileo-caecal valve. Metoclopramide is not effective for motion sickness.

Dosage and Administration

- Adults > 60kg: 10mg slowly IVI/IM
- Adults < 60kg: 5mg slowly IVI/IM

Adverse Effects

- CNS → Extra-pyramidal effects; depression
- GIT → Diarrhoea; abdominal cramps



Contra-indications

- Known allergy
- CNS → Epilepsy
- GIT → haemorrhage, obstruction, perforation or post-operative
- Children due to increased incidence of extra-pyramidal effects

Indications

Nausea and vomiting due to:

- Stimulation of CETZ by medication (eg. Morphine)
- Motility disorders of the GIT (eg. Gastro-enteritis)

Packaging

10mg/2ml ampoule

Half-life: 4-6 hours

Rocuronium

Nondepolarizing paralytic

Pharmacological Action

Rocuronium is a non-depolarizing paralytic. Nondepolarizing agents competitively block ACh at the nicotinic receptors. That is, they compete with ACh at the receptor without stimulating it. Thus, these drugs prevent depolarization of the muscle cell membrane and inhibit muscular contraction.



Packaging

10mg/ml in 2 ml ampoule

Indications

RSI Paralytic

Precautions

- Pregnancy & breastfeeding
- Hepatic impairment
- Elderly
- CVS Disease
- Neuromuscular disease
- Hypothermia
- Hypokalemia/ hypocalcemia/ hypermagnesia
- Dehydration
- Acidosis
- Hypercapnia

Onset of action: 1 min
Half-life: 45-60 mins

Adverse Effects

- CVS → tachycardia; hypotension
- Resp → bronchospasm; apnea; respiratory failure
- Anaphylaxis

Dosage and Administration

Adult and paediatric
RSI Paralytic: 0.6-1.2mg/kg IV
Maintenance: 0.1-0.2mg/kg continued infusion of 0.01mg/kg/min

Suxamethonium

Depolarizing paralytic

Onset of action: 30-60 sec
Duration of action: 3-5 min

Indications

RSI Paralytic

Pharmacological Action

Suxamethonium is the only depolarizing neuromuscular blocker (DNMB) which binds to nicotinic receptors at the neuromuscular junctions (NMJ) on the skeletal muscles to inhibit the normal functioning of the NMJ. Binding results in prolonged opening of nicotinic receptor ion channels (hence, fasciculations occur). Suxamethonium will bind for longer period compared to acetylcholine (Ach) because it is hydrolyzed by pseudocholinesterase instead of acetylcholinesterase in the plasma. Continued binding makes the receptor incapable of transmitting further impulses. Eventually, continuous depolarization disintegrate into gradual repolarization as the sodium channel closes or is blocked. Therefore, producing a muscle that is temporarily refractory to presynaptic further release of Ach and short-term flaccid skeletal muscle paralysis. Consciousness and pain sensation are not affected by Suxamethonium.



Packaging

100mg/2ml ampoule

Contra-indications

- History of malignant hyperthermia
- Pre-existing hyperkalaemia
- Known hypersensitivity
- 24 hours to 6 months history of major burn injury or para/tetraplegia: life-threatening hyperkalaemia may be caused

Adverse Effects

- Hypersensitivity
- Increased salivation, hypotension and bradycardia will probably result due to vagal stimulation and repeated doses

Dosage and Administration

<u>Adult</u>

- Initial 1-2 mg/kg (IV) injection. Customise subsequent doses according to patient's needs.
- 0.1% solution through IV infusion at 2-5 mg/minute for up to 1 hour.
- 3-4 mg/kg IM

<u>Paediatric</u>

- Under 1 year – 2 mg/kg, IV
- 1- 12 years – 1 mg/kg, IV

Sugammadex

Paralytic reversal agent

Pharmacological Action

Sugammadex is used to reverse neuromuscular blockade produced by the aminosteroid neuromuscular blocking drugs: rocuronium or vecuronium through encapsulation and inactivation of these muscle relaxants. It is a modified gamma-cyclodextrin, consisting of oligosaccharides linked around a central cavity. The muscle relaxant becomes entrapped within this cavity within a short time after sugammadex administration, neutralizing the relaxants, decreasing their plasma level and creating a concentration gradient between the neuromuscular end plate and plasma. This gradient causes displacement of the muscle relaxant from the end plate back into the plasma and further neutralization of the remaining relaxant.

Adverse Effects

- CVS → hypotension; bradycardia; prolonged QT interval
- Resp → Cough; bronchospasm
- GIT → Nausea, vomiting
- CNS → headache, insomnia

Indications

In the scenario of a large dose administration of rocuronium or vecuronium, when there is no evidence of any spontaneous recovery in the "cannot intubate-cannot ventilate" setting.

Precautions

- Pregnancy
- Breastfeeding
- Severe renal impairment
- Severe hepatic impairment
- Coagulopathy
- Hereditary vitamin K dependant clotting factor deficiencies
- Use in intensive care units
- Cardiovascular disease
- Sodium controlled diet

Dosage and Administration

<u>Adult</u>

16mg/kg IV bolus over 10 sec

<u>Children</u>

2mg/kg IV single dose

Packaging

Bridion 100 mg/mL
Solution for Injection



Onset of action: 2-3 mins
Half-life: 2 hours

Neostigmine

Anticholinesterase



Pharmacological Action

Neostigmine is a parasympathomimetic. The drug inhibits acetylcholinesterase which is responsible for the degradation of acetylcholine. As such, acetylcholinesterase is inhibited thereby increasing the concentration of acetylcholine. Neostigmine indirectly stimulates both nicotinic and muscarinic receptors which are involved in muscle contraction.

Contra-indications

- Mechanical gastrointestinal obstruction
- Urinary tract obstruction
- Intestinal obstruction
- Peritonitis

Precautions

- Asthma – may cause bronchosconstriction
- Bradycardia
- Cardiac arrhythmia
- Recent myocardial infarction
- Epilepsy
- Vagotonia
- Hyperthyroidism
- Peptic ulceration
- Parkinsonism
- Intestinal anastomoses
- Elderly patients – more susceptible to dysrhythmias

Adverse Effects

- Nausea
- Diarrhoea
- Increased salivation and sweating
- Abdominal cramps
- Bradycardia
- Bronchospasm
- Muscle cramps
- Hypotension

Onset of action: 20 mins
Half-life: 30 mins-2 hours

Dosage and Administration

Adult: 1-2.5mg IM/SC given at intervals to provide relief
Children: 0.01 – 0.04mg/kg

Indications

Reversal of neuromuscular blockade

Packaging

- 0.5mg/ml
- 2.5mg/ml

Calcium Chloride

Electrolyte

Pharmacological Action

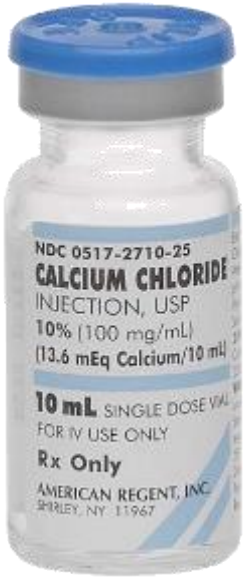
Functions in cell membrane stability, neurotransmitter release, hormone function, enzyme activation, nerve transmission, myocardial and smooth muscle function and blood clotting. Calcium is an essential mineral component of tissues and blood. Additionally, it plays a role in conduction of impulses in nerve and muscle contraction. Calcium has a positive inotropic effect on cardiac muscle Aids in general vasoconstriction. Calcium antagonizes the cardio-toxicity of hyperkalaemia by stabilising the cardiac cell membrane against undesirable depolarisation by restoring the normal gradient between threshold potential and resting membrane potential.

Indications

- Electrolyte imbalance: hypocalcaemia
- Hyperkalaemia associated with ECG changes (tall peaked T-waves, shortened QT interval, ST segment depression

Contra-indications

- Ventricular fibrillation
- Conditions associated with hypercalcaemia or hypercalciuria
- Elevated vitamin D levels



Precautions

- Renal impairment
- Hepatic impairment
- Hypocalcaemia of renal insufficiency
- Respiratory acidosis
- Respiratory failure
- Cardiac disease

Dosage and Administration

- Hyperkalemia: 1g IV as slow infusion of 1-2min in needed; max 8g/24hrs
- Magnesium toxicity: 500-1000mg IV daily over 2-5min, repeat if needed

Onset of action: 15 min
Half-life: Short

Adverse Effects

- CVS → Hyper/ hypotension; Bradycardia; arrhythmias
- CNS → Tingling sensation; sense of oppression
- Hot flushes

Packaging

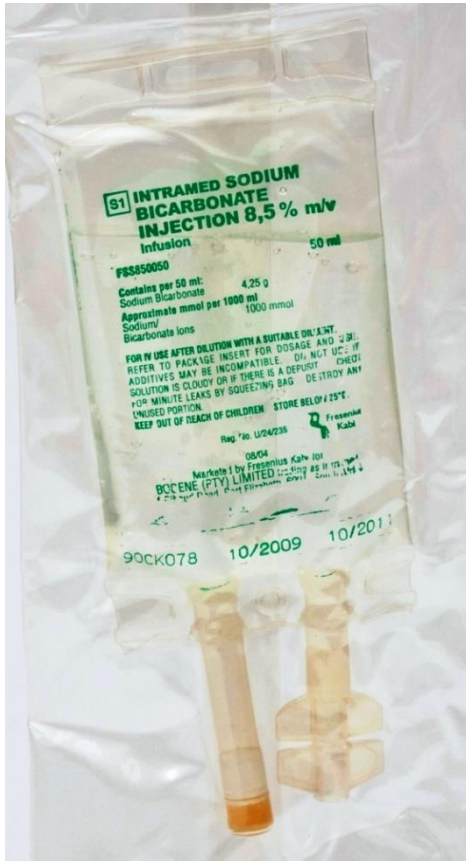
- 1g/10ml
- 100mg/ml

Sodium Bicarbonate

Mineral

Pharmacological Action

Sodium Bicarbonate is a mineral that dissociates into sodium and bicarbonate when it is administered. The molecule Bicarbonate (HCO3) then reacts/binds with H+ ions that gets converted into carbonic acid. This aid in the process of restoring the Bicarbonate-Carbonic Acid System, which can act as a buffer against acidosis.



Contra-indications

- Oliguric / anuric Renal Failure: Severe Renal impairment
- Decompensated Heart Failure: Congestive Heart Failure
- Hypernatremia
- Hypocalcemia
- Hypokalemia
- Hypochloremia
- Hypertension
- History of urinary calculi
- Metabolic and Respiratory alkalosis
- Oedema

Precautions

Never combine with calcium chloride in the same infusion and never administered via an infusion line with calcium containing solutions, unless the IV line is flushed with normal saline. Well placed and free-flowing line is mandatory.

Onset of action: 15 mins
Duration of action: 1-2 hour

Dosage and Administration

1ml/kg (8.5%) slowly IVI with free-flowing line

Indications

- Prior to release of prolonged entrapment to prevent acute crush syndrome
- ECG diagnosis of Suspected Hyperkalemia
- Cardiac arrest with known overdose and in Protacted PEA with suspected Hyperkalemia. (e.g. of overdoses, Tricyclics Antidepressants overdose)
- In cases of Tricyclic Antidepressant overdose associated with ECG abnormalities (QRS prolong > 100 msec), hemodynamic compromise, and malignant ventricular arrhythmia.

Adverse Effects

- CVS → hypervolemia; thrombophlebitis; aggravated CHF; decreased cardiac contractility
- CNS → Cerebral haemorrhage; tinnitus; hyper-irritability
- Resp → Pulmonary edema; hypercapnia
- Metabolic alkalosis
- Cellular dehydration

Dextrose 50%

Carbohydrate

Pharmacological Action

Glucose is a monosaccharide – the most basic unit to which all carbohydrates are broken down – and glucose is thus immediately available as a source of energy.

Packaging

- 20ml & 50ml ampoules of 50% solution
- 50ml vacolitre of 50% solution

Precautions

- Dehydration and hypovolemia – high concentrations of IV dextrose causes an increase in osmolarity that draws H2O from the cells and causes diuresis, aggravating dehydration.
- Intracranial haemorrhage – glucose leaking into the cerebral tissue will aggravate the injury and result in cerebral edema.

Indications

- Acute management of symptomatic hypoglycaemia – Blood glucose <3.5 mmol/L
- Decreased level of consciousness of unknown cause, with suspicion of associated hypoglycaemia



Adverse Effects

- Local vein irritation
- Thrombophlebitis
- Hyperosmolarity
- Diuresis
- Hyperglycemia

Dosage and Administration

<u>Adults</u>

10g (20ml of a 50% solution) slowly IV – Repeat every 5 min should blood glucose remain < 3.5mmol/L

<u>Children & Neonates</u>

1ml/kg of a 50% solution which is then diluted to a 12.5% solution with sterile water – Repeat every 5 min should blood glucose remain < 3.5mmol/L

Oral Glucose

Carbohydrate



Pharmacological Action

Glucose is a monosaccharide – the most basic unit to which all carbohydrates are broken down – and glucose is thus immediately available as a source of energy.

Packaging

- 25g and 50g powder sachets
- 25g and 50g gel sachets

Dosage and Administration

25g of gel applied to the oral mucosa of the patient with a gloved finger. Preferably dilute powder in glass of water if patient is conscious. Repeat every 5 min should blood glucose remain < 3.5mmol/L. Should patient be unconscious, liquid powder may be administered via NG tube, if no IV access or glucagon IM is available.

Indications

- Acute management of symptomatic hypoglycaemia – Blood glucose <3.5 mmol/L

Precautions

- Patient must be lateral if unconscious
- Avoid Aspiration

Adverse Effects

- Hyperglycemia

Glucagon

Hyperglycemic agent

Pharmacological Action

A biosynthetic form of glucagon (pancreatic hormone) which releases glucose from the liver by means of glycogenolysis, thereby increasing blood glucose levels. Glucagon inhibits GIT motility for about 15 mins; it increases bile flow and decreases secretion of digestive enzymes. It has positive inotropic and positive chronotropic cardiac effects produced by direct adenylate cyclase stimulation, bypassing the beta-adrenoceptors. It thus enhances myocardial performance by increasing cAMP concentrations in a manner identical to that of catecholamines, but probably acting via its own receptor.

Contra-indications

- Phaeochromocytoma
- Must not be administered to patients who are hypersensitive to glucagon.
- History of allergy to beef or porcine protein
- Safety in pregnancy & lactation has not been established.

Dosage and Administration

Adults

- 1mg IMI / IVI / SC for hypoglycaemia - IMI is preferred route
- 3mg IV initially as adjuvant treatment of calcium channel blocker or beta blocker OD, followed by 3mg/hour infusion
- Severe anaphylactic reactions: 1 - 2 mg every 5 min IM or slow IV **if** unresponsive to adrenaline, & especially if on beta blockers

Paediatrics

- Hypoglycemia: < 20kg (< 6 years) : 0.5mg IMI / IVI / SC; > 20 kg : 1mg IMI / IVI / SC
- Severe anaphylaxis: < 20µg/kg IM or slow IV (max 1 mg) **if** unresponsive to adrenaline, & especially if on beta blockers

Adverse Effects

- Nausea & vomiting may occur – dose related (may be caused by the hypoglycaemia)
- Uncommonly, dizziness, light-headedness, skin rash, dyspnea
- Hypokalaemia
- Myocardial ischaemia may be aggravated in patients with pre-existing cardiac disease



Indications

- Acute management of symptomatic hypoglycaemia (blood glucose < 3.5mmol/L) Only if adequate IV access or IV dextrose is not available or effective.
- Severe anaphylactic reactions if patient is unresponsive to adrenaline, and especially if the patient is taking beta blockers.
- Severe symptomatic bradycardia from beta blockade overdose not responding to other medications e.g. adrenaline
- Calcium channel blocker OD

Precautions

- 1mg / vial freeze-dried glucagon plus syringe containing 1ml H2O for injection
- Reconstitute with provided solution: Inject 1ml H2O for injection into vial, shake to dissolve, then draw up solution. Do NOT mix with saline.

Thiamine

Vitamin

Pharmacological Action

Thiamine is an important co-enzyme in cellular carbohydrate metabolism. In the absence of sufficient thiamine, some neuronal cells are vulnerable to ischaemia and necrosis, leading to Wernicke's encephalopathy and Korsakoff's psychosis.

Packaging

200mg/2ml ampoule



Indications

- Suspected Wernicke's encephalopathy.
- Prior to intravenous administration of 50% glucose, where the possibility of thiamine deficiency is suspected, e.g alcoholism, malnourishment

Adverse Effects

- Anaphylactic reactions have been reported but are rare.
- Pruritus and dermal flushing with large doses.

Dosage and Administration

- Adults: 100mg IVI bolus, slowly
- Children: 25 – 50mg IVI bolus, slowly

Activated Charcoal

Carbon

Pharmacological Action

Activated charcoal adsorbs many poisonous compounds to its surface, thereby reducing their absorption by the GIT

Contra-indications

- SHOULD NOT BE USED IN POISONING WITH iron, organophosphates, ethanol, lithium, boric acid, cyanide, ethylene glycol, methanol, petroleum products, strong acids and alkalis
- Unprotected airway in a patient with decreased level of consciousness
- Do not use if the container was not properly sealed (de-activation due to moisture exposure)

Indications

To assist in the treatment of certain cases of overdoses and poisonings where the agent/s was/were orally ingested – within first hour of ingestion

Precautions

Patients with a decreased level of consciousness need to be intubated before activated charcoal can be administered via a nasogastric tube

Dosage and Administration

Adult and Paediatric: 0.5g/kg - 1g/kg mixed with water, given orally or administered via the nasogastric tube

Adverse Effects

Mild constipation

Packaging

Fine black powder in bottles of 25g and 50g



Flumazenil

Benzodiazepine Antagonist



Pharmacological Action

Binds to GABA receptors (competing with benzodiazepines). Flumazenil is a benzodiazepine antagonist that specifically blocks the central effects of agents acting through the benzodiazepine-receptor complex by competitive inhibition. Has no agonist activity

Precautions

- Antagonises the anticonvulsant effects of benzodiazepines.
- Risk of withdrawal syndrome in benzodiazepine dependant patients, possibly resulting in seizures.
- The half-life is shorter than that of most benzodiazepines **(therefore monitor for recurrent sedation and respiratory depression).**
- Pregnancy: excretion in milk unknown. Human studies not available. Animal studies show risk.
- Alcoholism
- Head Trauma

Contra-indications

- Known hypersensitivity
- Suspected Tricyclic Anti-depressant overdose
- Unknown mixed-drug overdose
- Use in undifferentiated coma confers risk.
- Patients with a high risk of convulsions
- Neonates
- Patients receiving benzodiazepines for life-threatening conditions (ICP control, status epilepticus)

Indications

- Reversal of CNS sedative effects of benzodiazepines in anaesthesia or in intensive care situations or inadvertent/self-administered benzodiazepine overdoses
- Reversal of central nervous system sedative effects and respiratory depression due to iatrogenic benzodiazepines administration.

Adverse Effects

- CVS → Flushed skin; thrombophlebitis; arrhythmias; hypertension; chest pain
- CNS → Excitation; convulsions
- Acute Benzo withdrawal in dependent patients
- Tremors & involuntary movements

Onset of action: 1-2 mins
Duration of action: 45 mins

Packaging

- 0.1mg/1ml 10ml ampoule = 1.0mg
- 0.1mg/1ml 5ml ampoule = 0.5mg

Dosage and Administration

<u>Adults:</u>

0.2mg slowly IVI over 30sec into free-flowing IV → Repeat 0.1mg every 1 min if no response seen; Max dose: 1mg (Usual range = 0.3-0.6mg)

<u>Children:</u>

0.01mg/kg bolus dose over 15 sec → may repeat after 45 sec then every min; Max dose: 0.05mg/kg

Naloxone

Narcotic Antagonist

Pharmacological Action

Naloxone competes with narcotic drug's opiate receptors in the central nervous system to displace the narcotic analgesics from their receptor sites. It will thus reverse the effects of narcotic analgesics such as respiratory depression, stupor, etc., but it also has the ability, because of its action, to cause acute withdrawal in a patient who is dependent on narcotics

Indications

The only therapeutic goal is to reverse any respiratory depression in a suspected narcotic overdose, and not to fully awaken such patients, who may become violent should acute withdrawal occur

- Only to be administered after adequate oxygenation and ventilation
- Reversal of respiratory depression due to acute opiate usage
- Neonatal respiratory depression secondary to the administration of opioids to the mother in the previous 4 hours

Precautions

- Suspected narcotic dependence (may precipitate an acute withdrawal syndrome) – administer IV very slowly to reverse respiratory depression only.
- The effect of naloxone is usually shorter than that of many long acting narcotics and therefore repeated doses may have to be given in order to maintain the desired effect; Continued monitoring and observation of such patients is mandatory.
- Provide adequate ventilation until respiratory depression has been adequately reversed

Dosage and Administration

<u>Adults:</u>

0.4mg slowly IVI (0.1mg/kg) → Repeat every 5 mins, up to 2mg; Should 2mg fail to elicit the desired response, then overdose with agents other than opioids should be considered

<u>Children:</u>

0.1mg/kg slowly IVI/ IMI → Repeat every 5 mins, up to max of 2mg



Onset of action: 2 mins
Duration of action: 45 mins

Packaging

0.4mg/ml ampoule

Contra-indications

- Known hypersensitivity / allergy
- Infants of opioid addicted mothers

Adverse Effects

- Acute withdrawal symptoms in opioid dependent patients → Sweating, piloerection, tremors; Agitation and convulsions
- Mydriasis
- Excessive lacrimation
- Nausea and vomiting
- CVS → Ventricular tachycardia or fibrillation, hypotension or hypertension

Promethazine

Antihistamine



Pharmacological Action

Promethazine is a non-selective antihistamine with considerable activity at other receptor sites: it combines potent H1-antagonism with antiemetic, anticholinergic, and marked sedative activities, and some hypotensive effects. The role of promethazine in acute anaphylaxis is adjunctive to other core resuscitative approaches.

Indications

Acute anaphylaxis/severe allergy

Onset of action: 3-5 mins
Duration of action: 12 hours

Precautions

- Epilepsy
- Cardiac disease
- Asthma
- Narrow-angle glaucoma
- Elderly patients – more susceptible to CNS and anticholinergic effects.

Dosage and Administration

<u>Adults</u>

25mg deep IM or slowly IVI after 10-fold dilution with sterile water

<u>Children:</u>

0.5mg/kg

- >12 yrs: 25mg IM or slow IV
- 6-12 years: 12.5mg IM or slow IV
- 2-5 years: 6.25mg IM or slow IV

Adverse Effects

- CNS → drowsiness; dizziness; fatigue; incoordination
- Paradoxical stimulation → seizures; hallucinations; nervousness
- GIT → nausea; vomiting; epigastric pain
- Anticholinergic effects → dry mouth; blurred vision; urinary retention

Packaging

- 25mg/ml and 50mg/2ml ampoules
- Oral antihistamines: 10mg and 25mg tablets

Contra-indications

- Not recommended for children under 2 years of age.
- Known hypersensitivity / allergy to promethazine